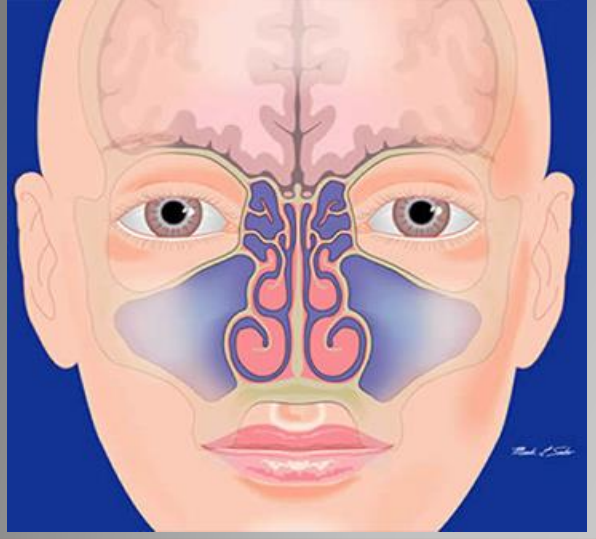


Radiological Anatomy of Head and Neck Disease



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Year 3
Module A

Learning Objectives

I. Anatomy the nasal cavity and paranasal sinuses.
II. Acute sinusitis.
III. Acute epiglottises

I. Nasal and Paranasal sinuses

1. Nose (external).

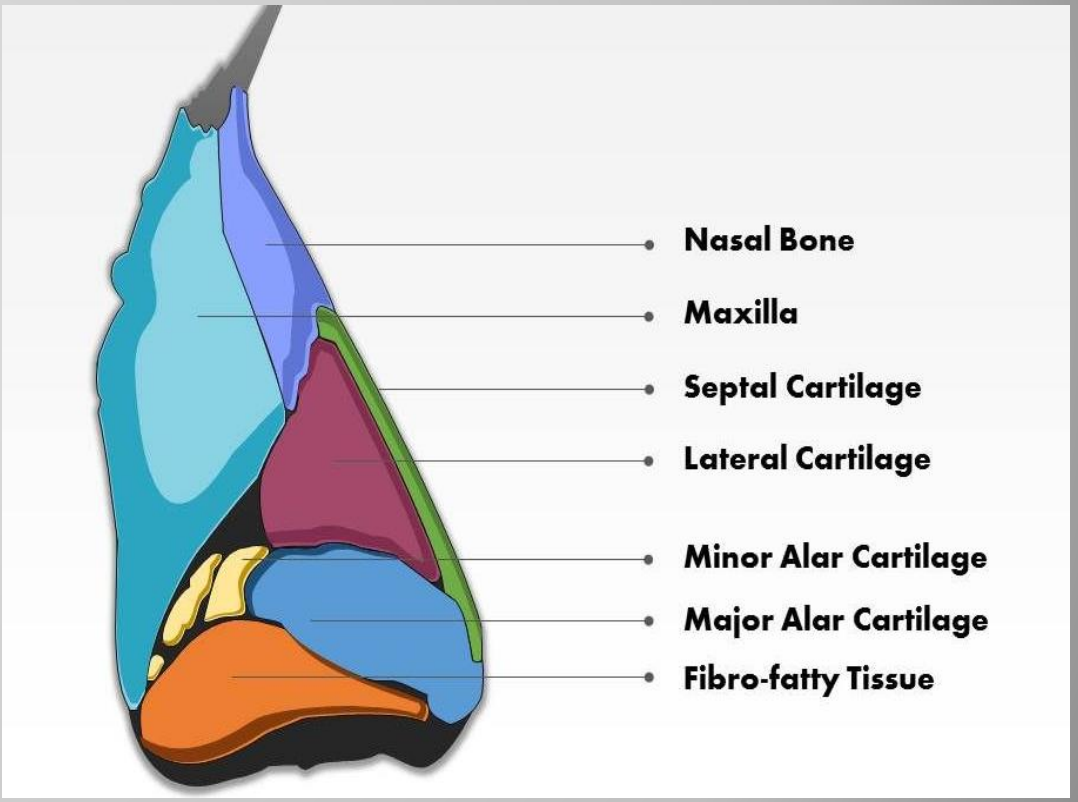
A. Two paired of bony structures:

i. Nasal bones

ii. Maxillary bones

B. Covered by cartilages

C. Fibro-fatty tissue



I. Nasal and Paranasal sinuses

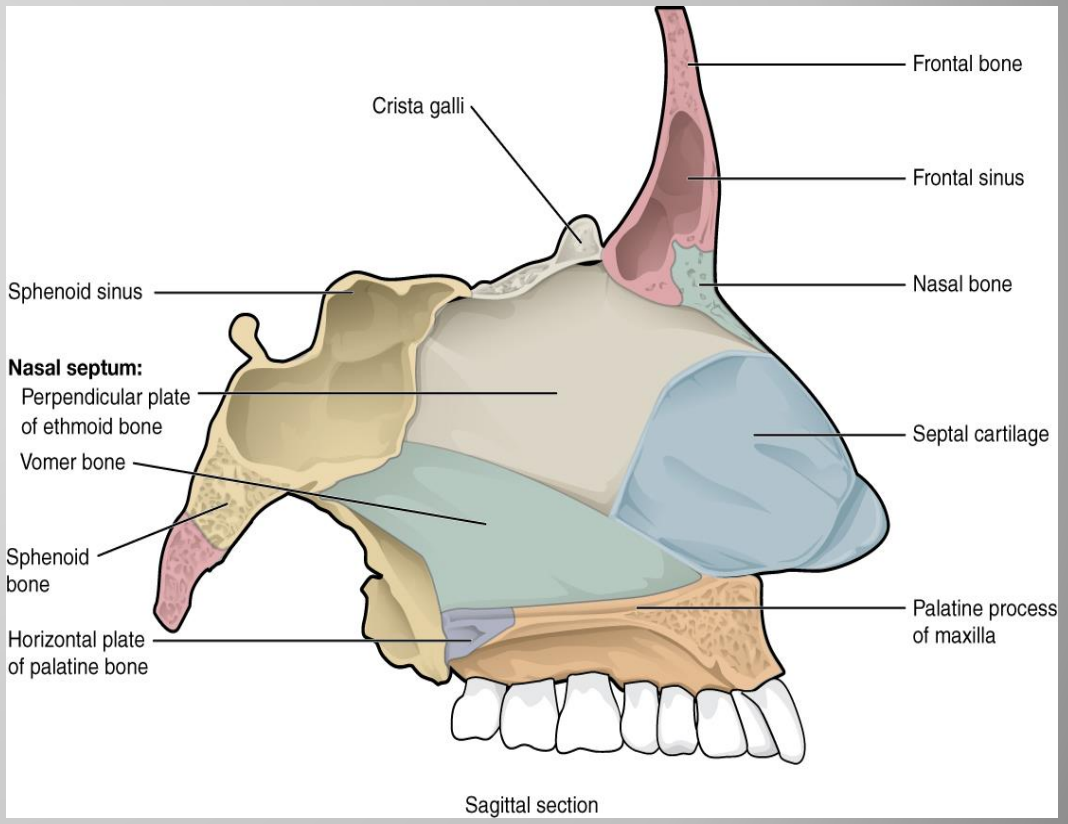
2. Nasal septum (Midline)

A. Bony septum (Posterior):

I. Vomer

II. Perpendicular plate of ethmoid bone

B. Cartilaginous septum (Anterior)

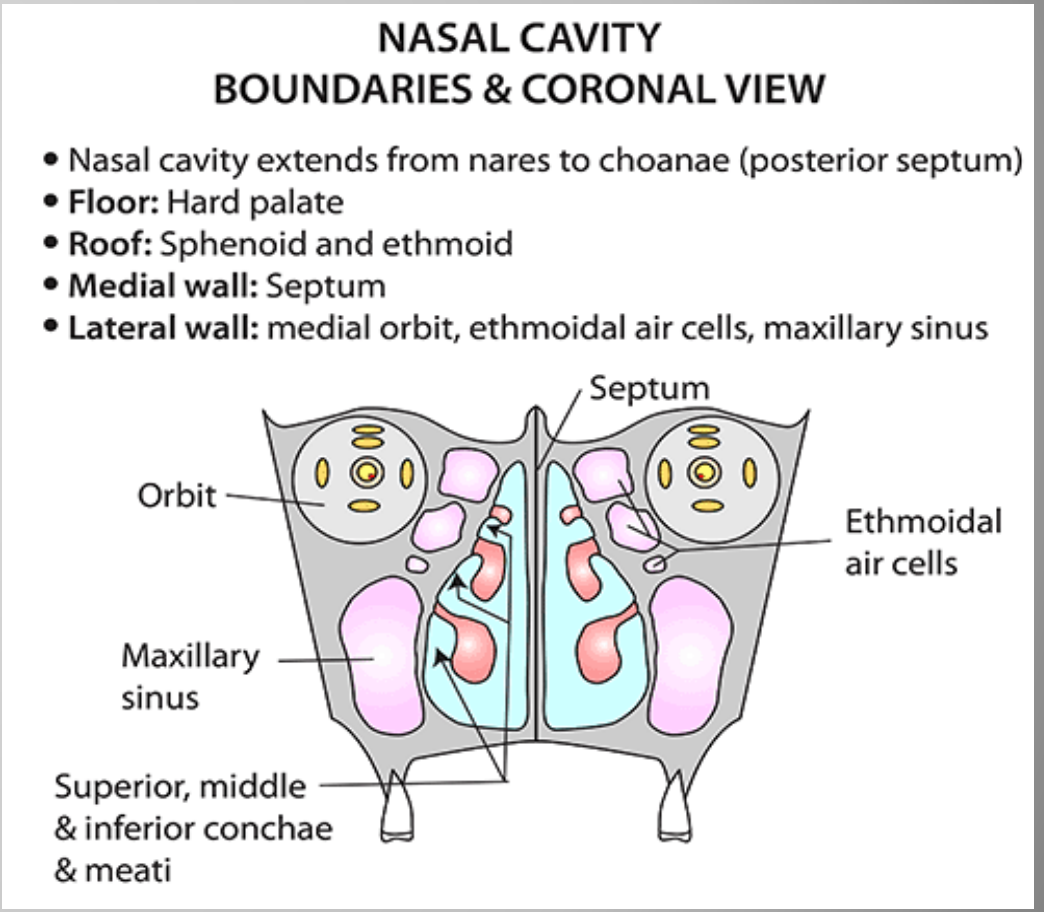


Sagittal section

I. Nasal and Paranasal sinuses

3. Nasal cavity (Internal)

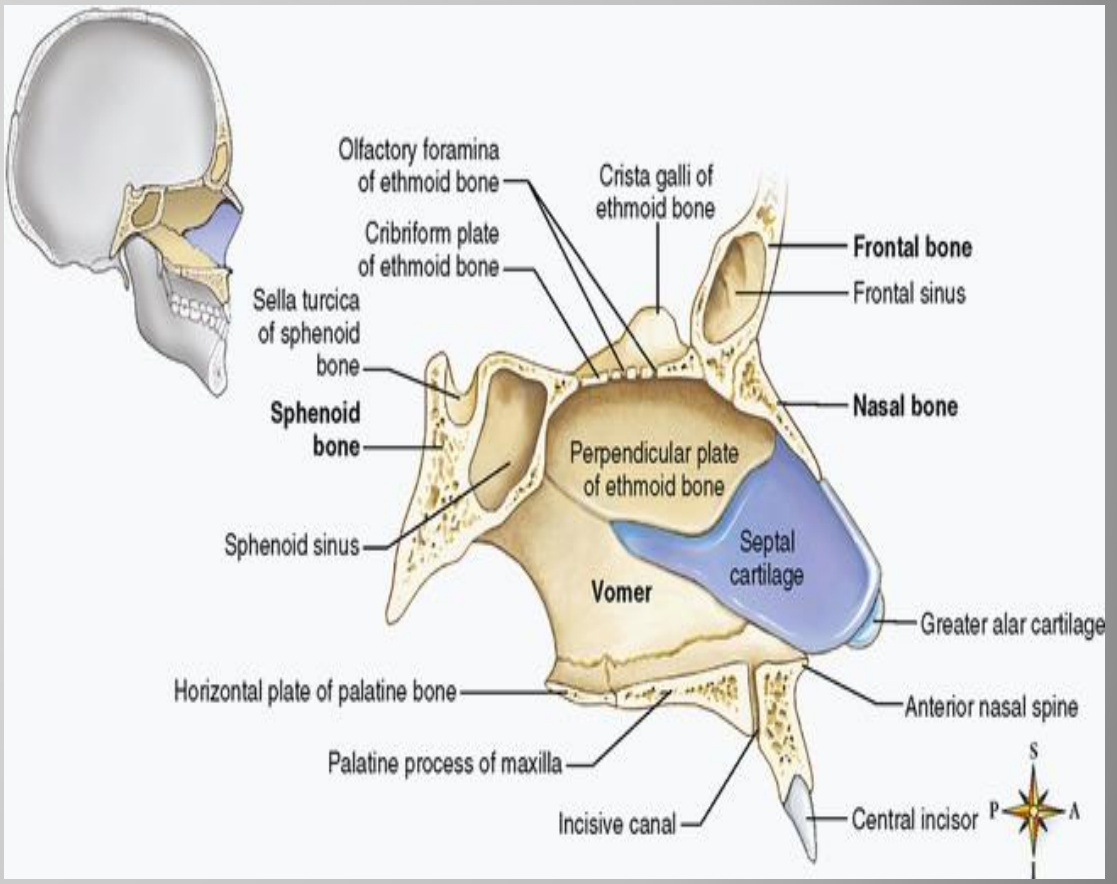
- **Superiorly = Sphenoid bone and Cribriform plate of the ethmoid bone which pass the sensory fibers of the Olfactory nerve (1st Cranial nerve)**
- **Inferiorly = Maxillary bone and hard palate**
- **Medially = Nasal septum**
- **Laterally = maxillary sinus wall and medial wall of the orbital ring (Lamina papyracea)**



I. Nasal and Paranasal sinuses

3. Nasal cavity (Internal)

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I. Nasal and Paranasal sinuses

• 3. Nasal cavity (Internal)

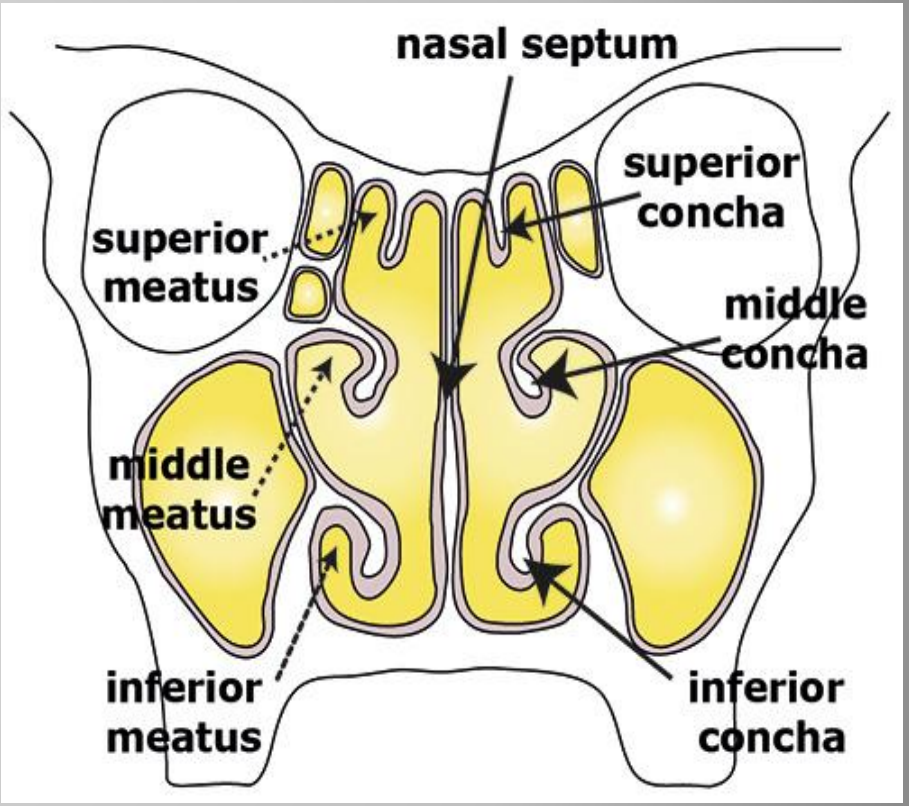
Each nasal cavity has the following structures:

- **Anterior opening (Nostril) and Posterior opening (Choanae/posterior Nostril).**

- **Three Turbinates (conchae) arranged as Superior, middle and inferior.**

***The main function of the Turbinates are filtration, heating, and humidification of inhaled air.**

- **Below each Turbinate is a corresponding Meatus.**



I. Nasal and Paranasal sinuses

- **3. Nasal cavity (Internal)**

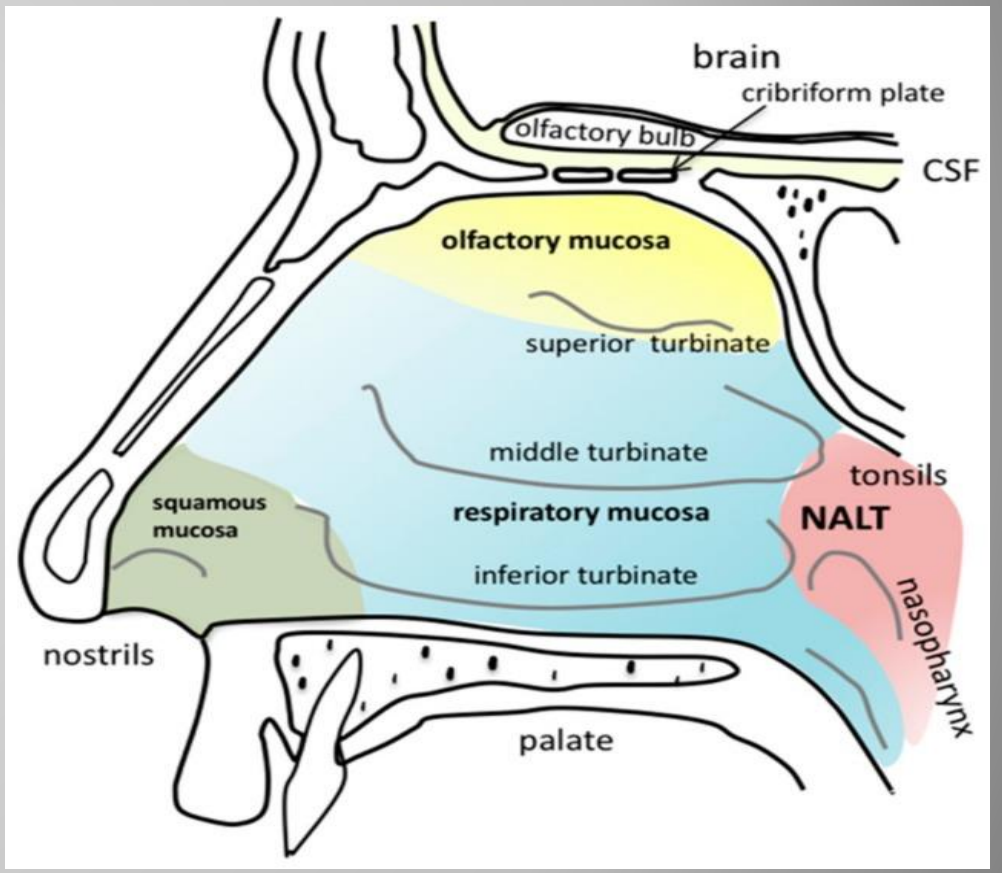
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I. Nasal and Paranasal sinuses

- **4. Paranasal Sinuses**
  - **Consist of 4 paired air-filled spaces. They are named for the facial bones in which they are located:**

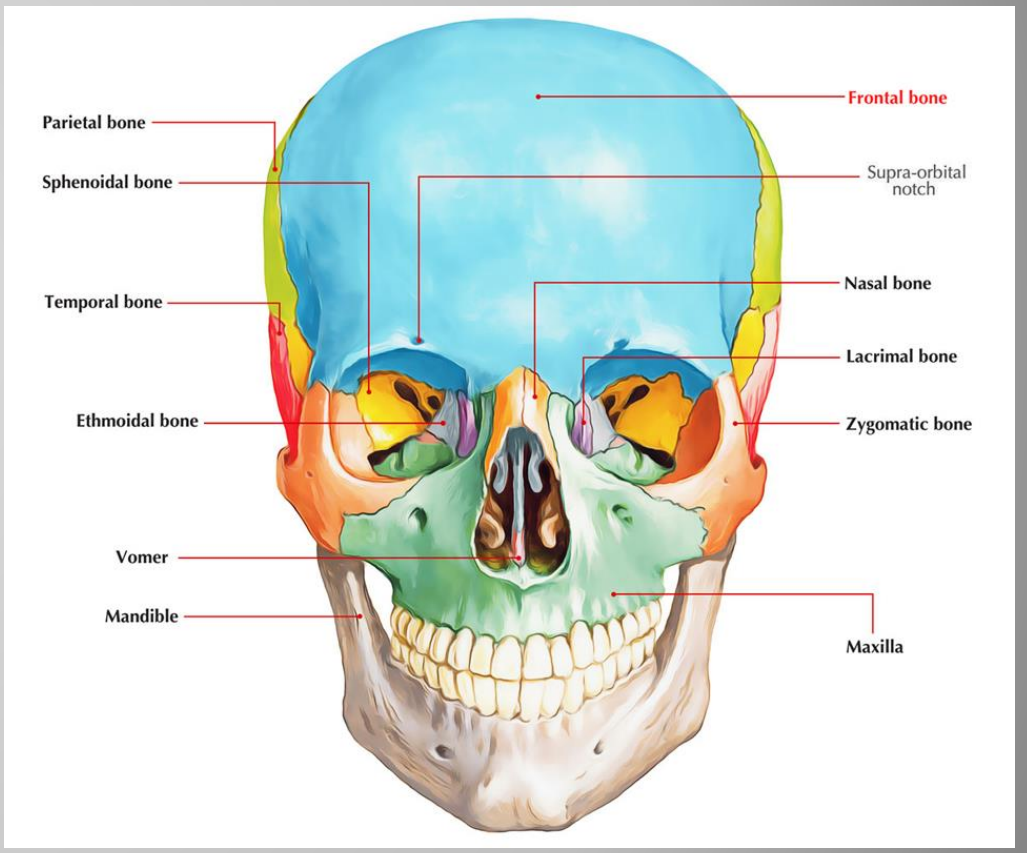
I. **Maxillary sinus**

II. **Sphenoid sinus**

III. **Ethmoid sinus (Anterior and Posterior)**

IV. **Frontal sinus**

***The main functions of the paranasal sinuses are lightning the skull, acts as resonance champers during speech, regulate intranasal pressure and humidifying/warming inspired air.**



I. Nasal and Paranasal sinuses

- ## 4. Paranasal Sinuses

- **Consist of 4 paired air-filled spaces. They are named for the facial bones in which they are located:**

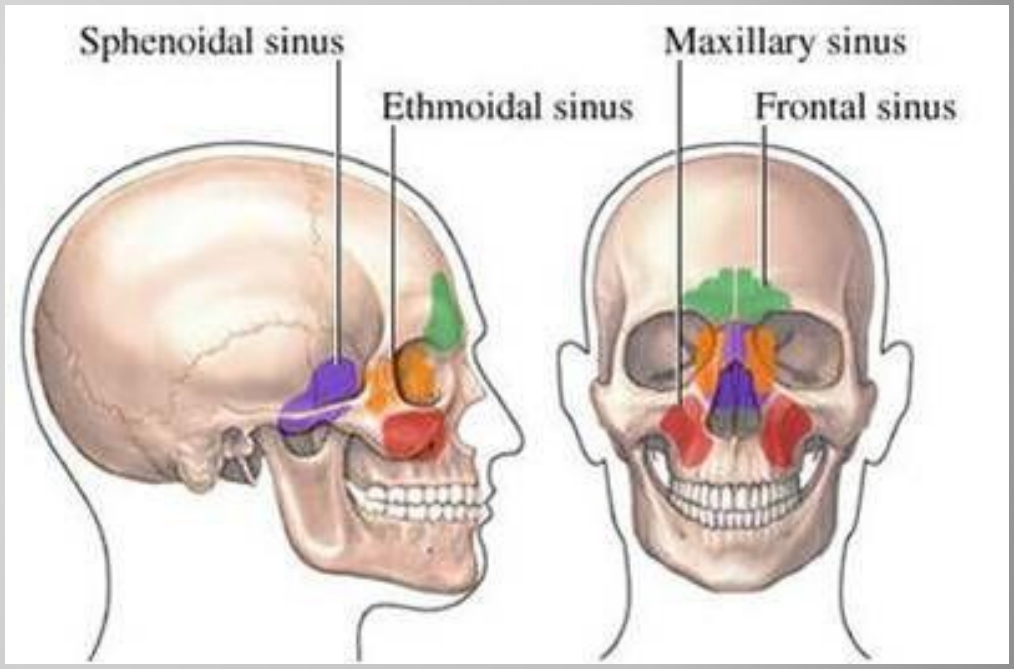
I. **Maxillary sinus**

II. **Sphenoid sinus**

III. **Ethmoid sinus (Anterior and Posterior)**

IV. **Frontal sinus**

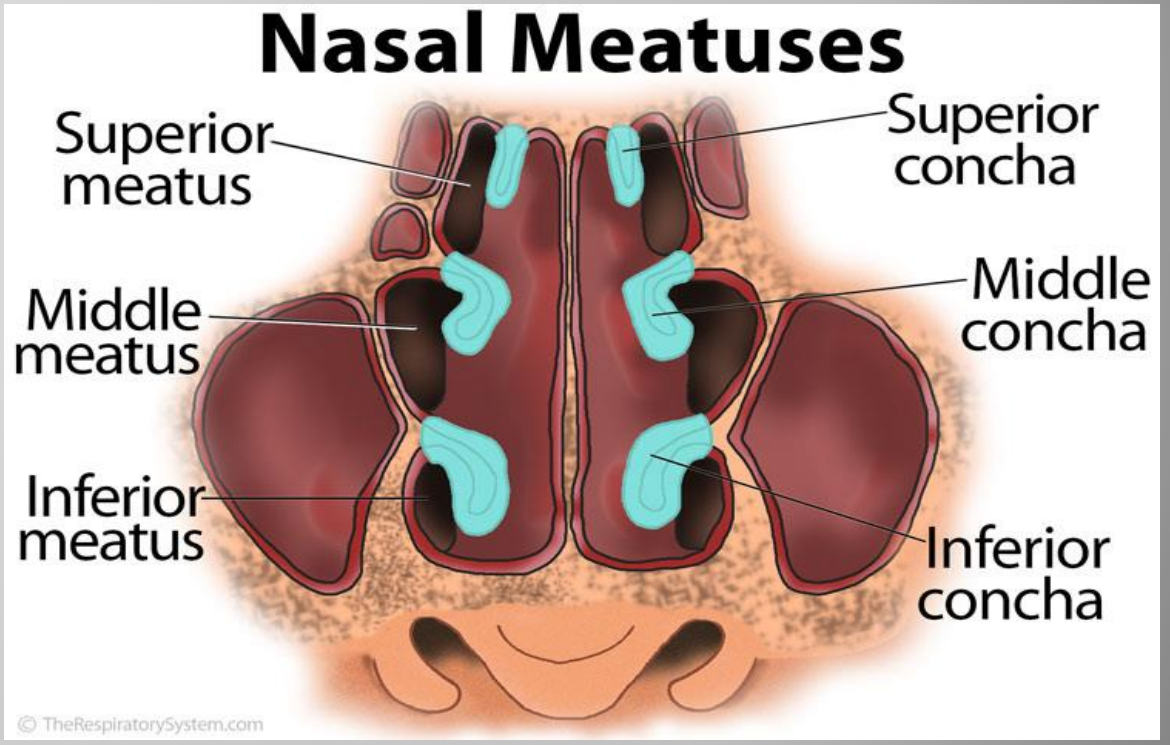
- **The main functions of the paranasal sinuses area lightning the skull, acts as resonance champers during speech, regulate intranasal pressure and humidifying/warming inspired air.**



I. Nasal and Paranasal sinuses

* **Meatus : Located inferior to each nasal Turbinate (conchae) and they receive drainage of the paranasal sinuses and nasolacrimal duct.**

Superior	**Sphenoid and posterior ethmoid sinuses**
Middle	**Frontal, anterior ethmoid and Maxillary sinuses**
Inferior	**Nasolacrimal duct**



I. Nasal and Paranasal sinuses

* Development of paranasal sinuses

Sinus	Status at Birth	Age at which Adult size achieved
Maxillary sinus	present	15 years
Ethmoid sinus	present	12 years
Frontal sinus	absent	Post pubertal
Sphenoid sinus	absent	14 years

I. Nasal and Paranasal sinuses

- **Imaging Modality**

I. X-ray
II. CT
III. MRI



X-ray

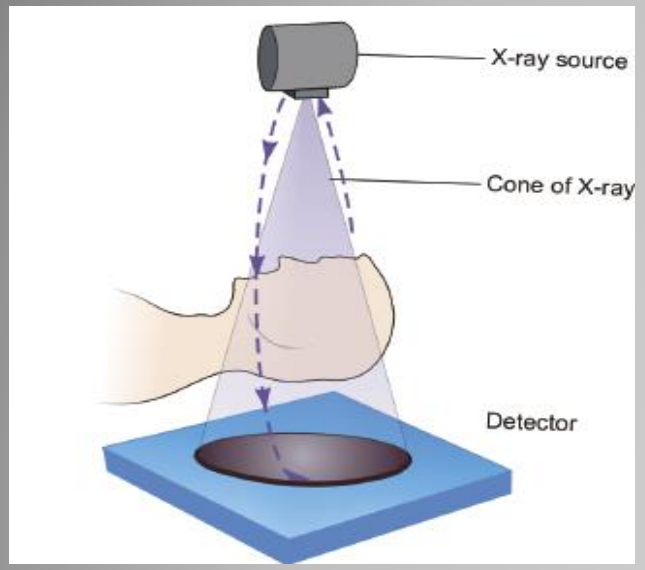


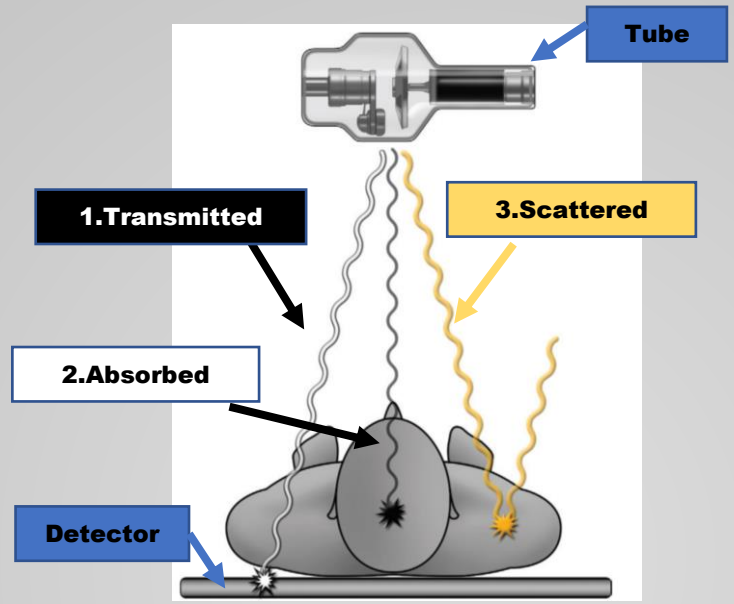
CT



MRI

X-ray (Plain film) Approach





***Three Types of Radiation During an Imaging Procedure:**

1. **Transmitted Radiation:** It passed directly to the detector because it Doesn't interact with patient (about 10 %).
2. **Absorbed Radiation:** It interacts with the tissues of the patient.
3. **Scatter Radiation:** It is not transmitted or absorbed by the patient (changes the pathway).

TABLE 1-1 FIVE BASIC DENSITIES SEEN ON CONVENTIONAL RADIOGRAPHY

Density	Appearance
Air	Absorbs the least x-ray and appears "blackest" on conventional radiographs
Fat	Gray, somewhat darker (blacker) than soft tissue
Fluid or soft tissue	Both fluid (e.g., blood) and soft tissue (e.g., muscle) have the same densities on conventional radiographs
Calcium	The most dense, naturally occurring material (e.g., bones); absorbs most x-rays
Metal	Usually absorbs all x-rays and appears the "whitest" (e.g., bullets, barium)

AIR

FAT

Soft tissue

Bone

Metallic

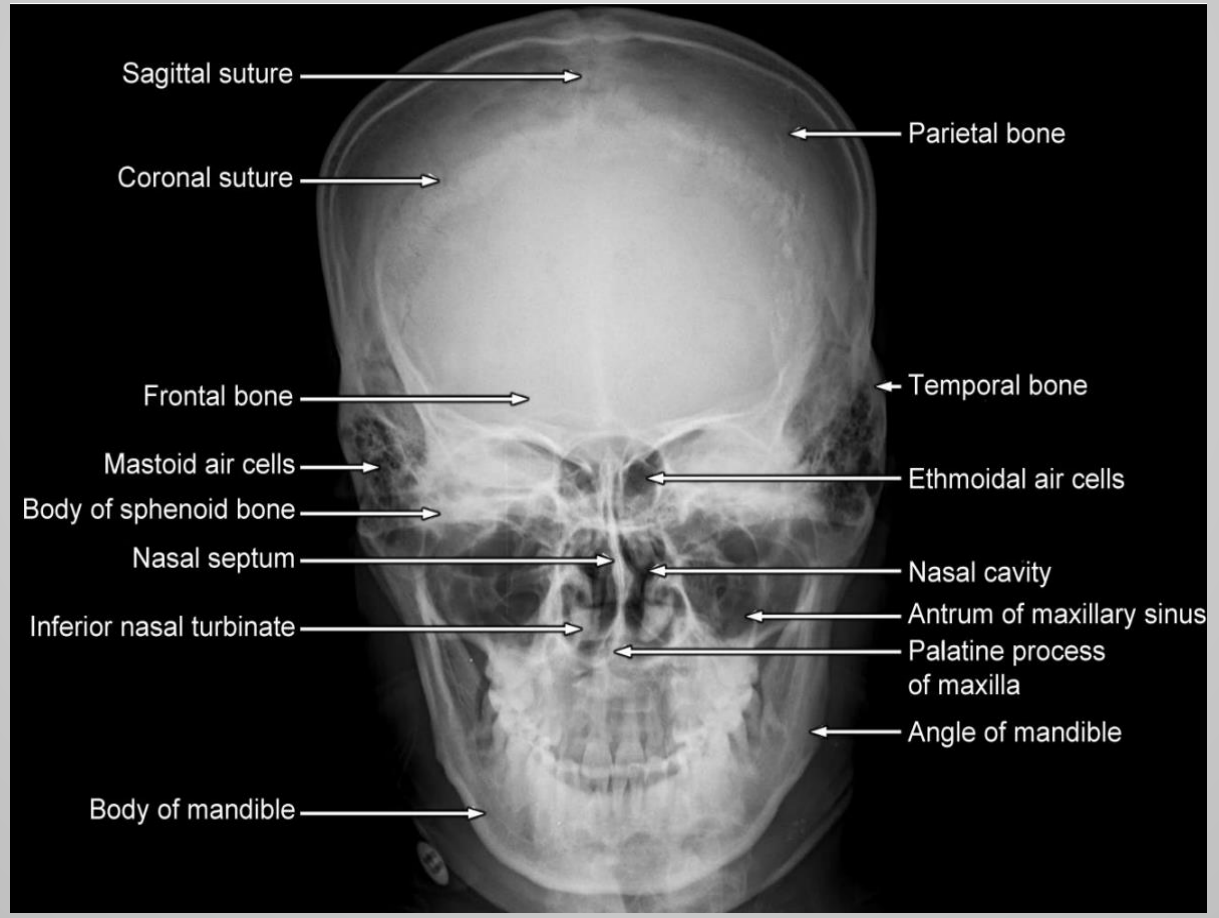
Increased soft tissue density → **Increased Absorbed Radiation**

eTABLE 1-4 WHITE AND BLACK: TERMS FOR EACH MODALITY

Modality	Terms Used for "White"	Terms Used for "Black"
Conventional radiographs	Increased density Opaque	Decreased density Lucent
CT	Increased (high) attenuation Hyperintense Hyperdense	Decreased (low) attenuation Hypodense

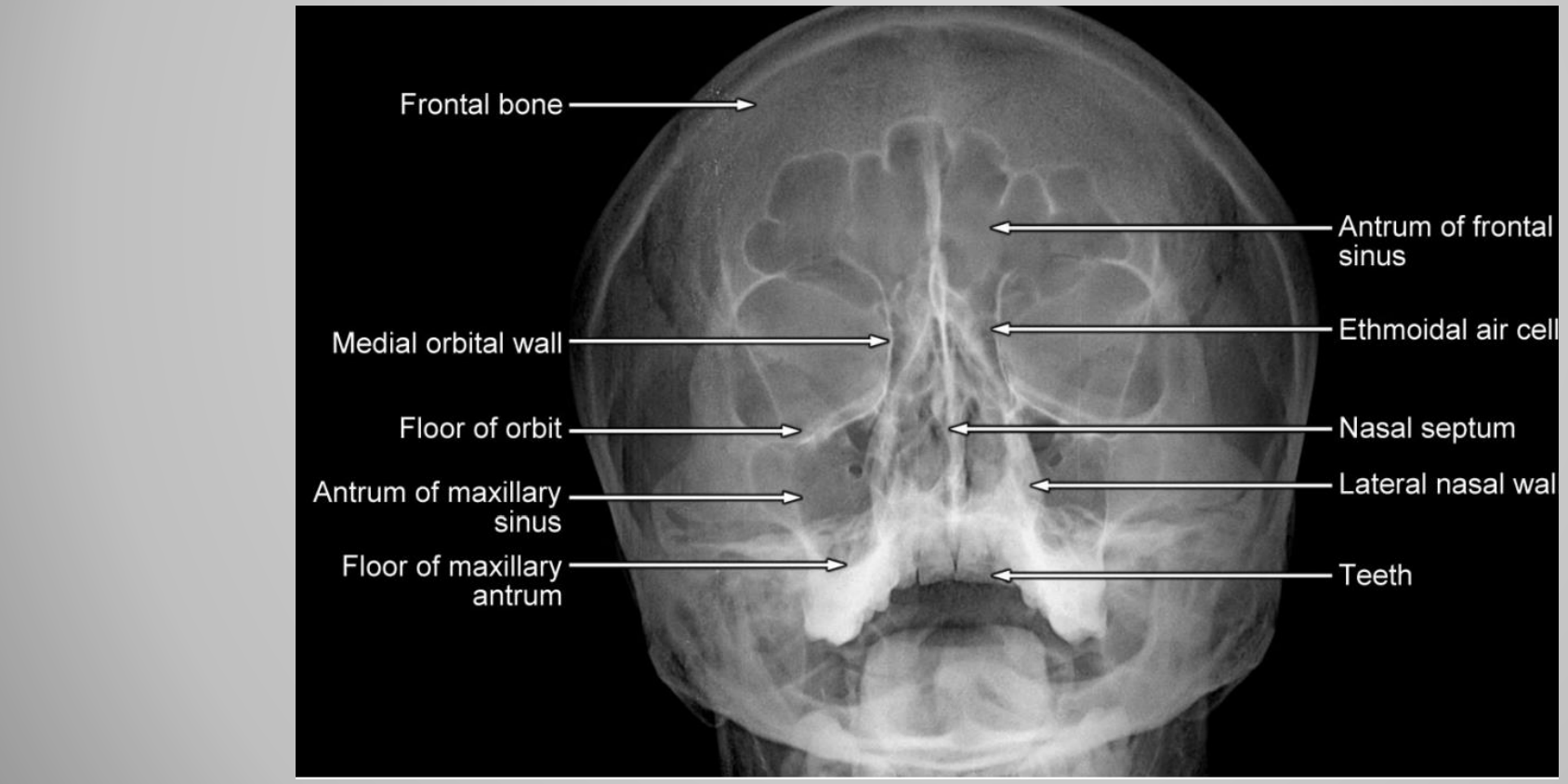
I. Nasal and Paranasal sinuses

I. X-ray (Caldwell view)



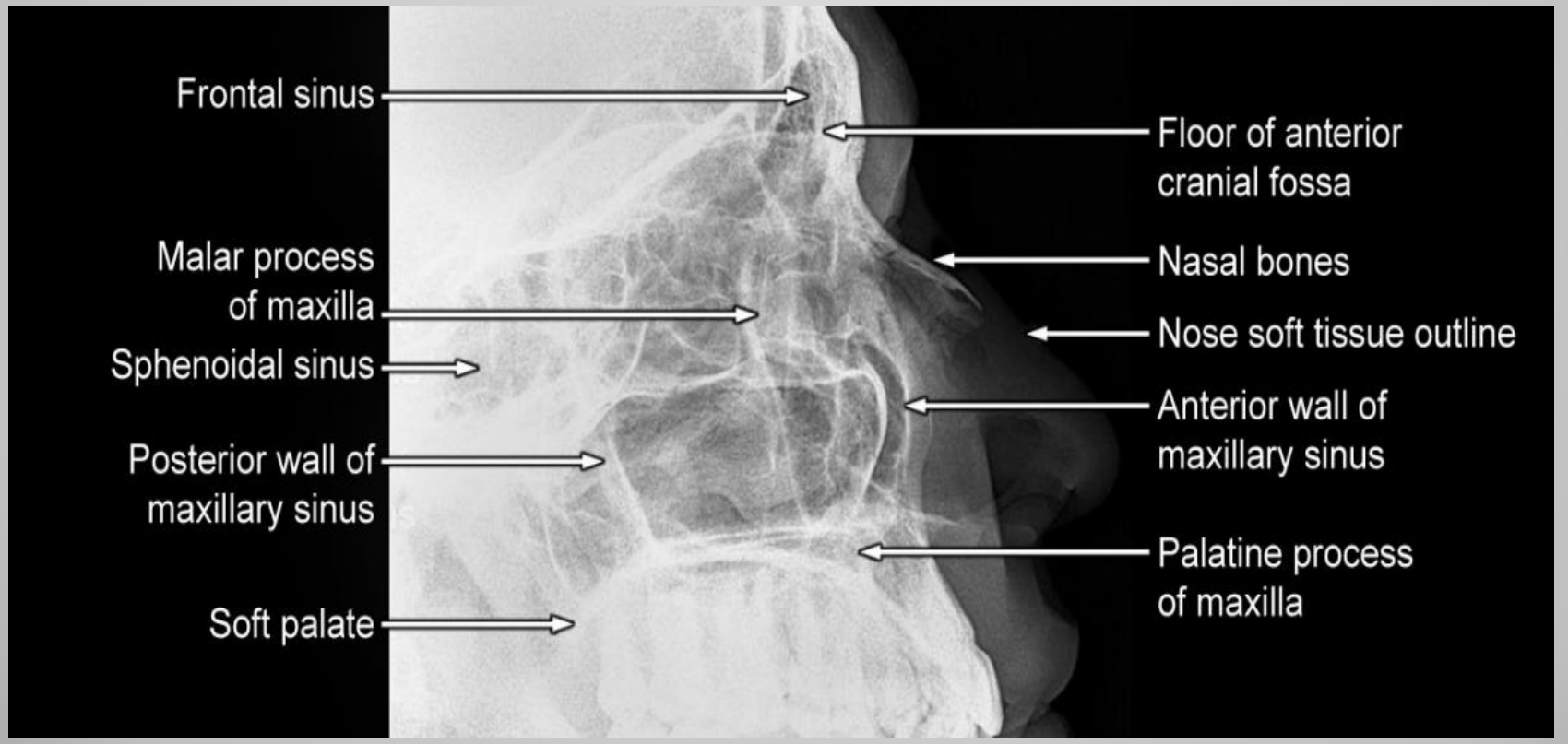
I. Nasal and Paranasal sinuses

I. X-ray (Water's view)



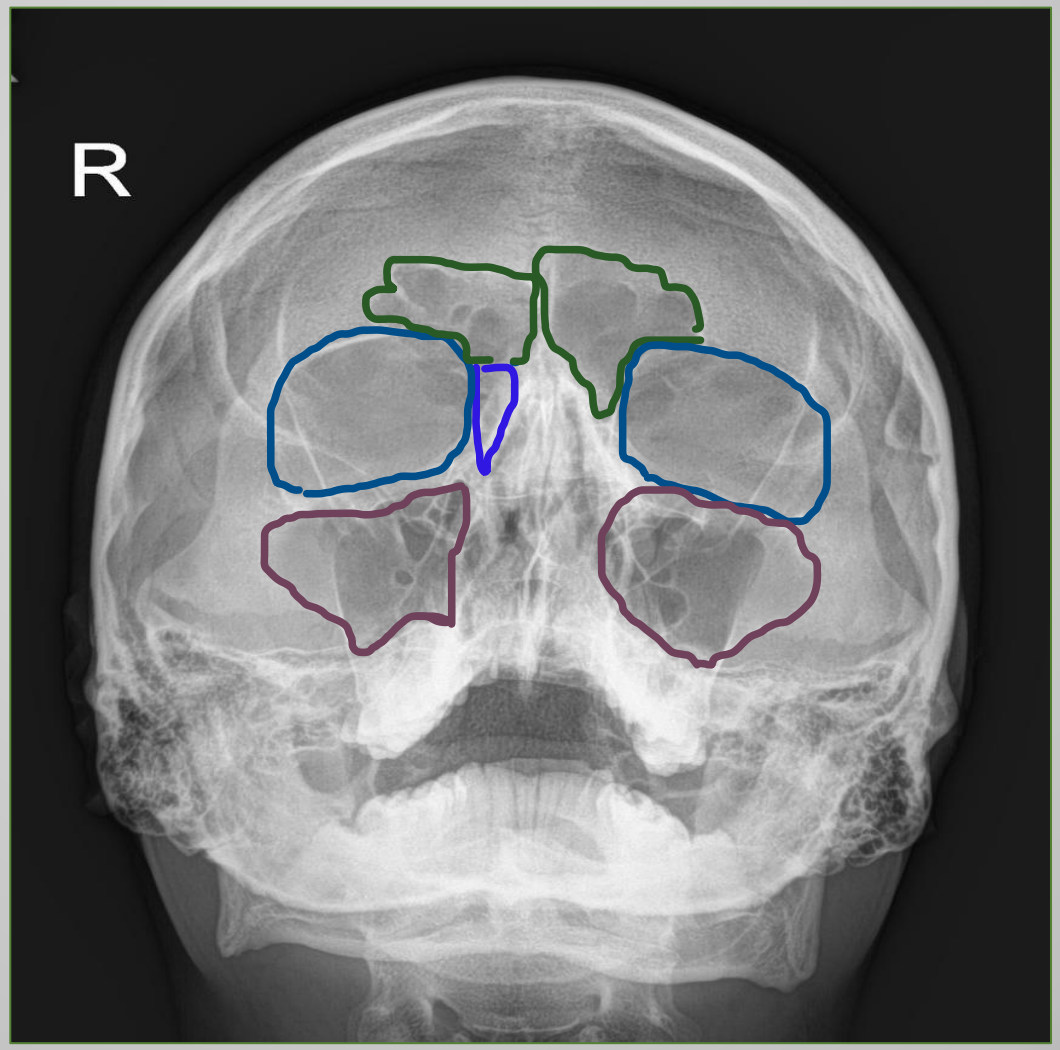
I. Nasal and Paranasal sinuses

I. X-ray (Nasal bone view)



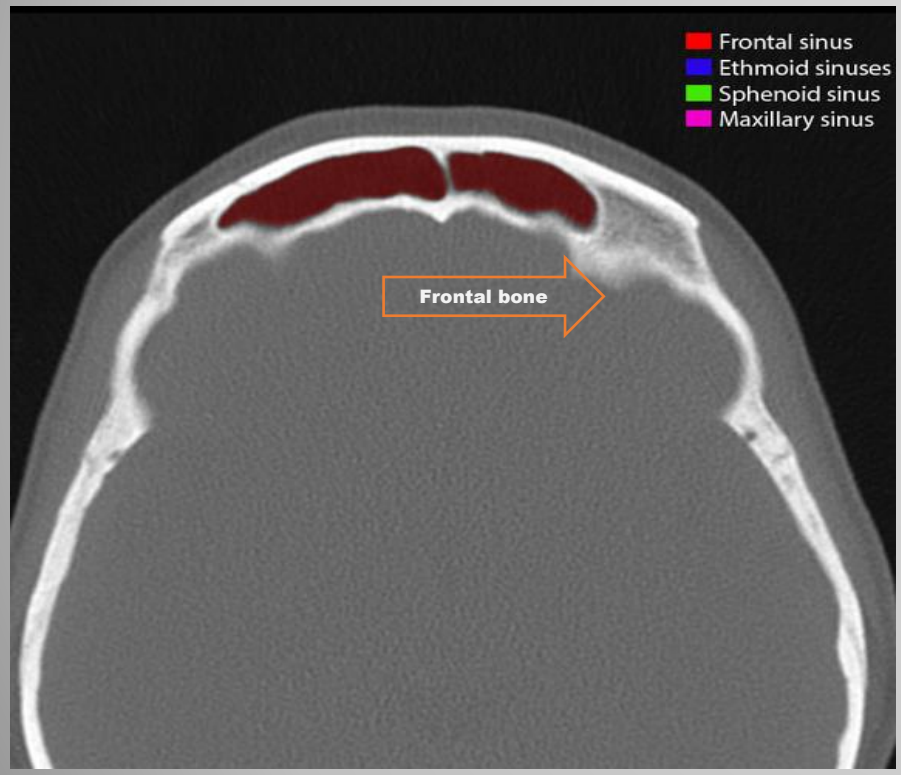
I. Nasal and Paranasal sinuses

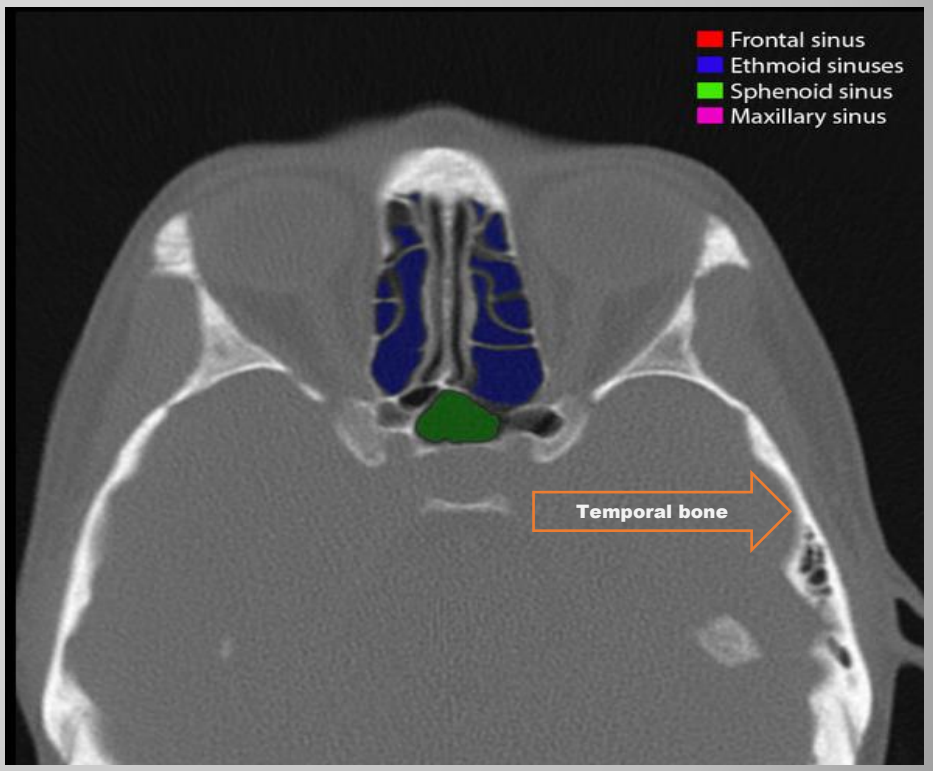
I. X-ray (??)



I. Nasal and Paranasal sinuses

- **CT (Axial View)**

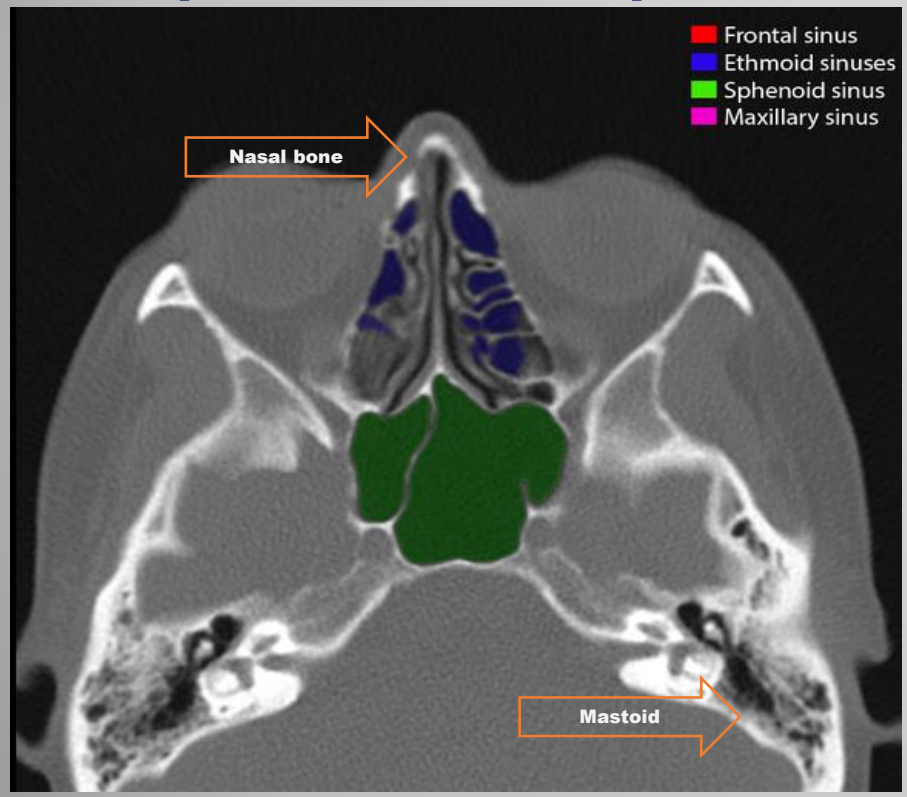


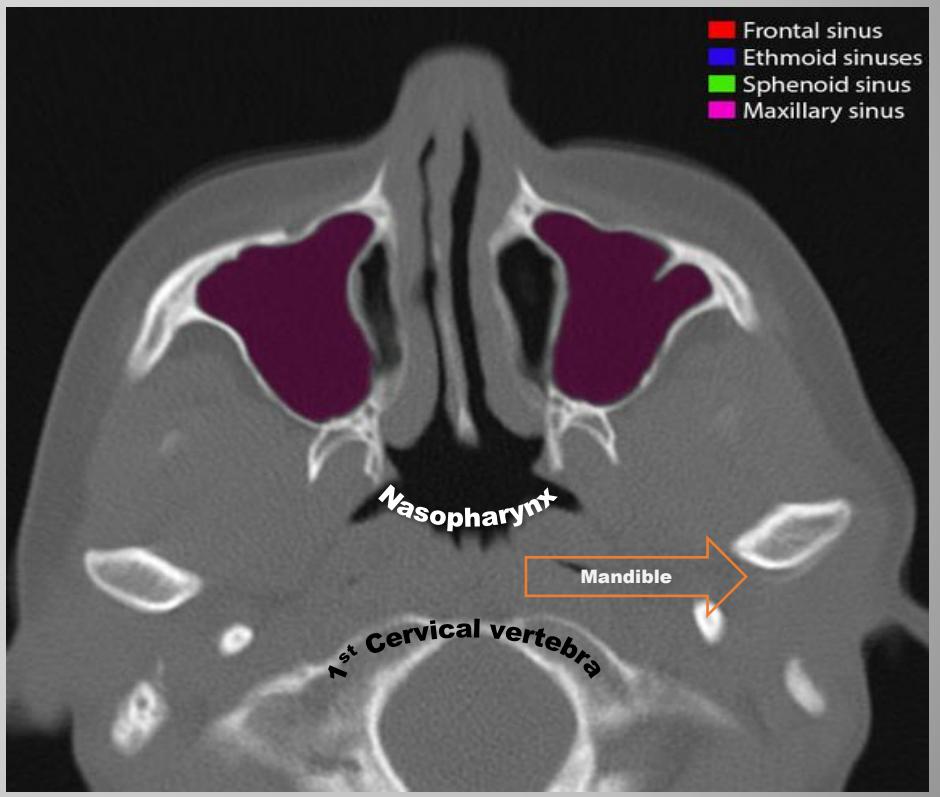


Superior → Inferior

I. Nasal and Paranasal sinuses

- **CT (Axial View)**

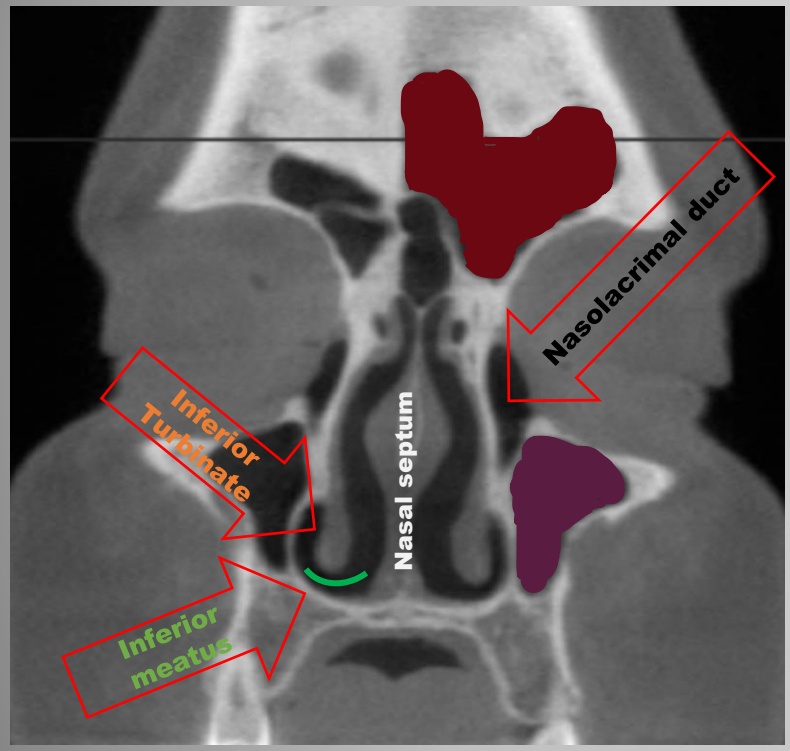


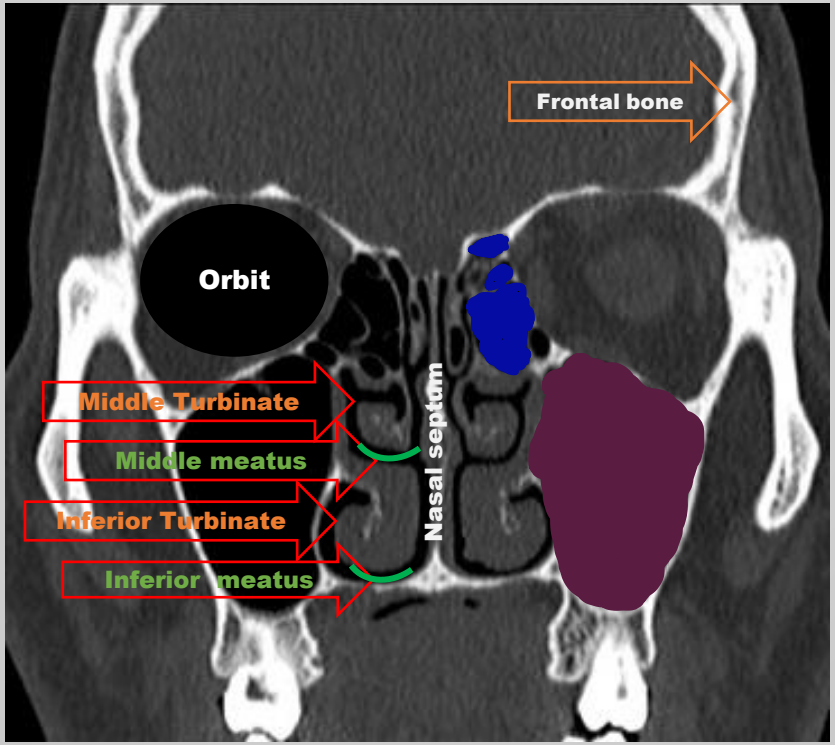


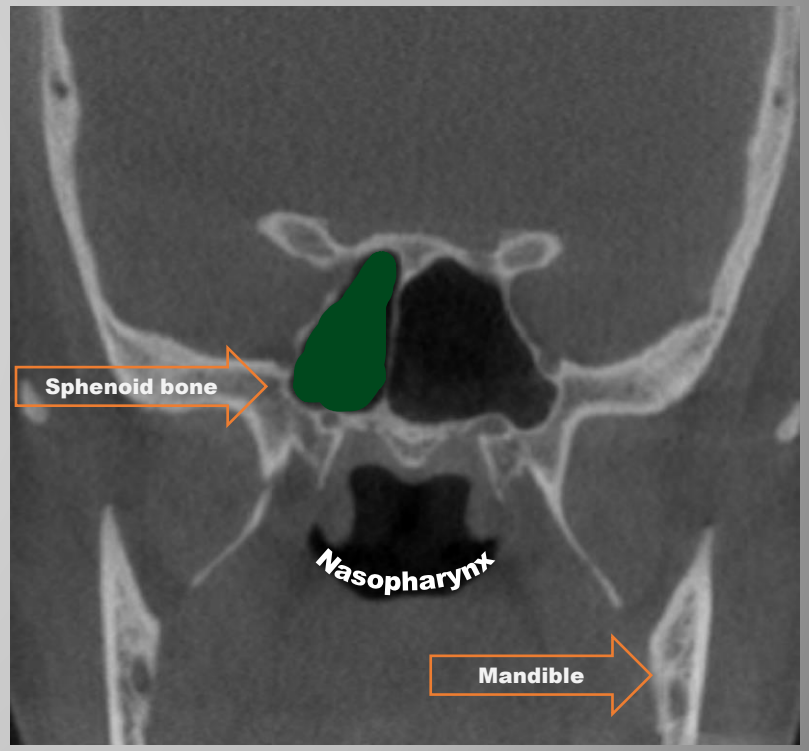
Superior → Inferior

I. Nasal and Paranasal sinuses

- CT (Coronal View)



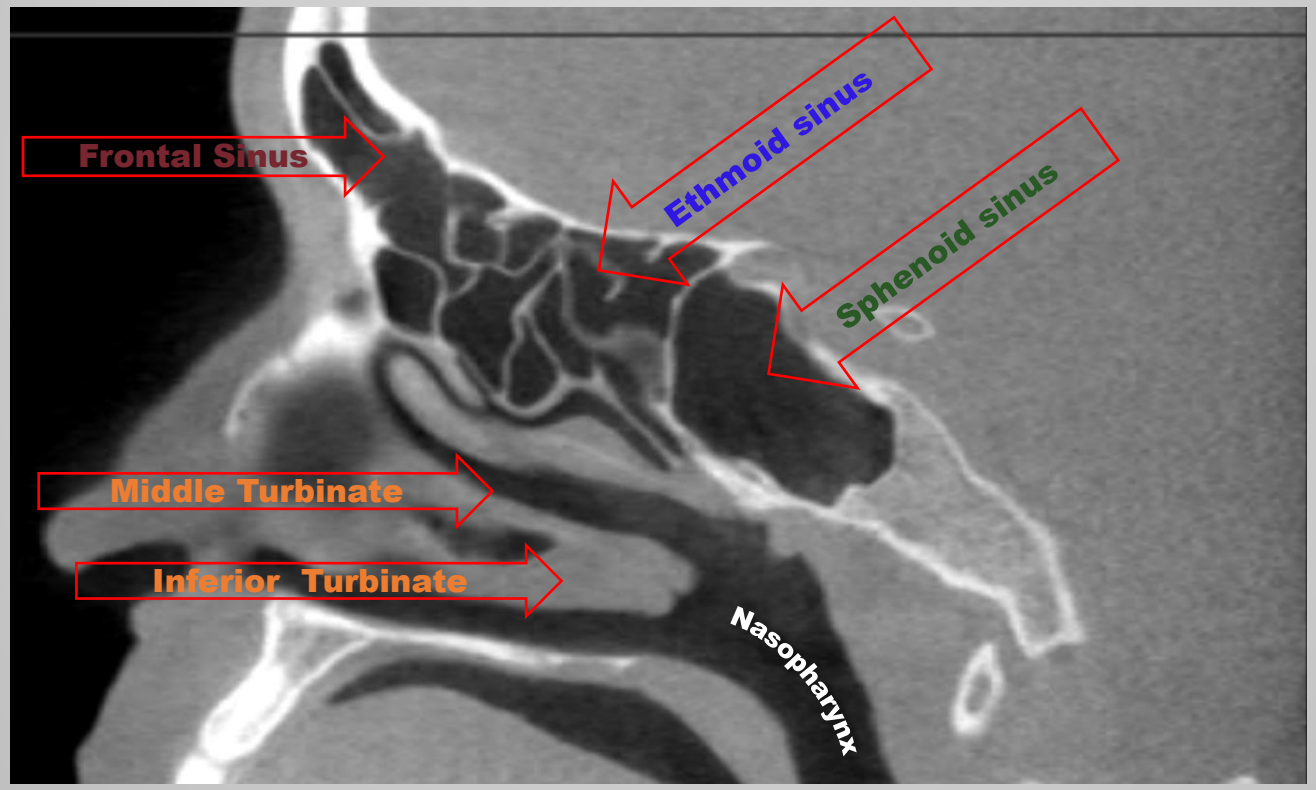




Anterior → Posterior

I. Nasal and Paranasal sinuses

- CT (Sagittal View)



I. Nasal and Paranasal sinuses

- **MRI**



I. Nasal and Paranasal sinuses

- **Imaging Modality**

	X-Ray	CT	MRI
Acceptability	↑↑↑	↑↑	↑
Indication	**Acute sinusitis**	1. Persistent Acute Sinusitis 2. Extension of the Disease (Acute sinusitis) 3. Trauma 4. Pre and Post Surgery assessment	1. Complicated Acute sinusitis 2. Immunocompromised patient with sinusitis 2.Tumor
Radiation	↑↑	↑↑↑	**No Radiation**
Cost	↑	↑↑	↑↑↑

II. Acute Sinusitis

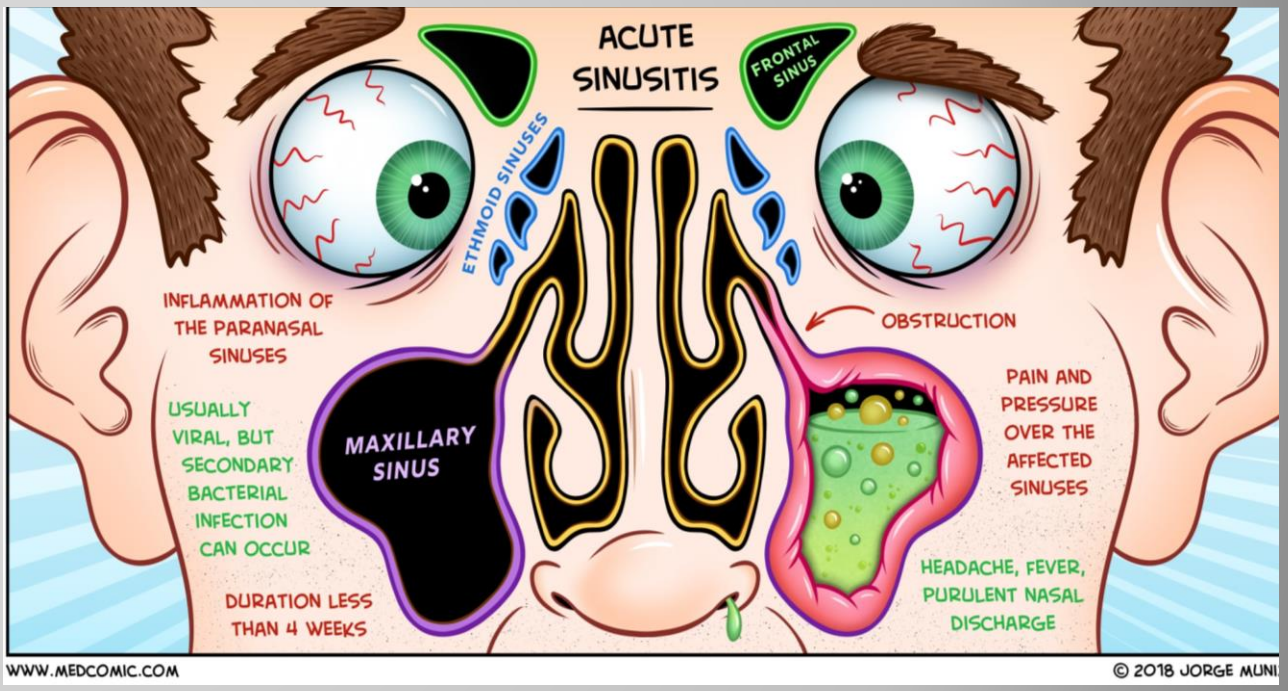
- **Definition:**

Acute inflammation of the paranasal sinus mucosa that lasts less than 4 weeks.

- **Etiology:**

1. Usually following a viral upper respiratory tract infection.

2. Complicated Dental cares.



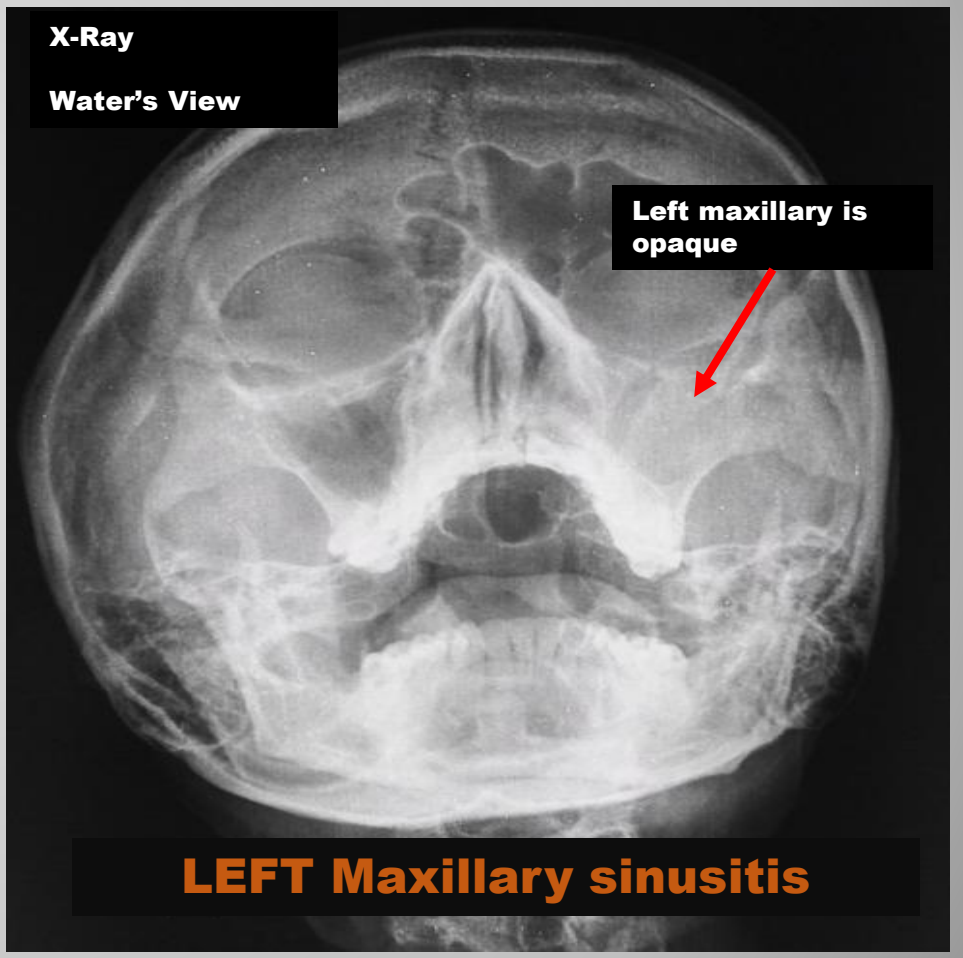
II. Acute Sinusitis

- **Clinical presentation:**

1. PureInt Nasal Discharge.
2. Cough
3. Low Grade Fever
4. Nasal congestion
5. Headche
6. Frontal and Maxillary Tenderness on palpation
7. Eye pain

- **Risk factors:**

1. Allergy in children.
2. Long standing intubation.



II. Acute Sinusitis

- **Radiological Manifestation:**

I. **X-ray (Plain film): Opacification of the sinuses and gas-fluid level best seen in the maxillary sinus.**

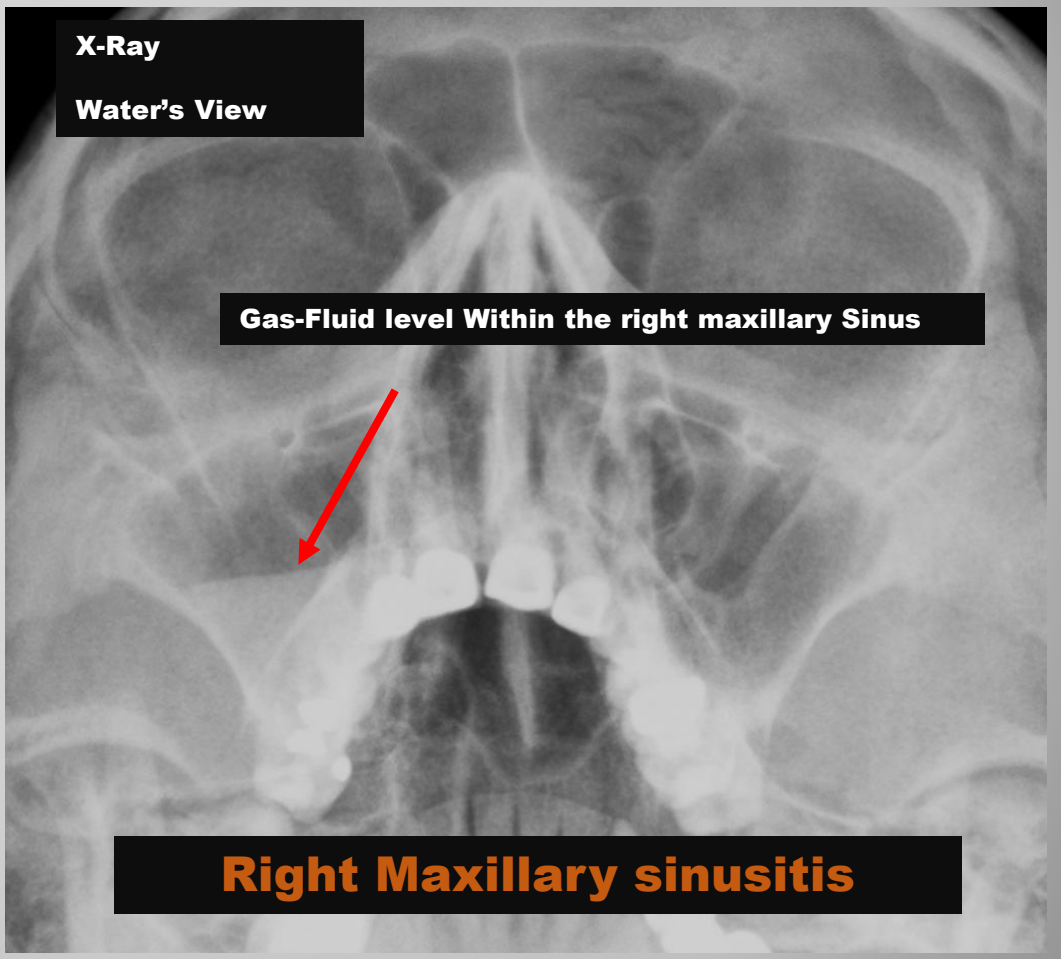
II. **Compute Tomography (CT):**

Better anatomical delineation and assessment of inflammation extension, causes and complications.

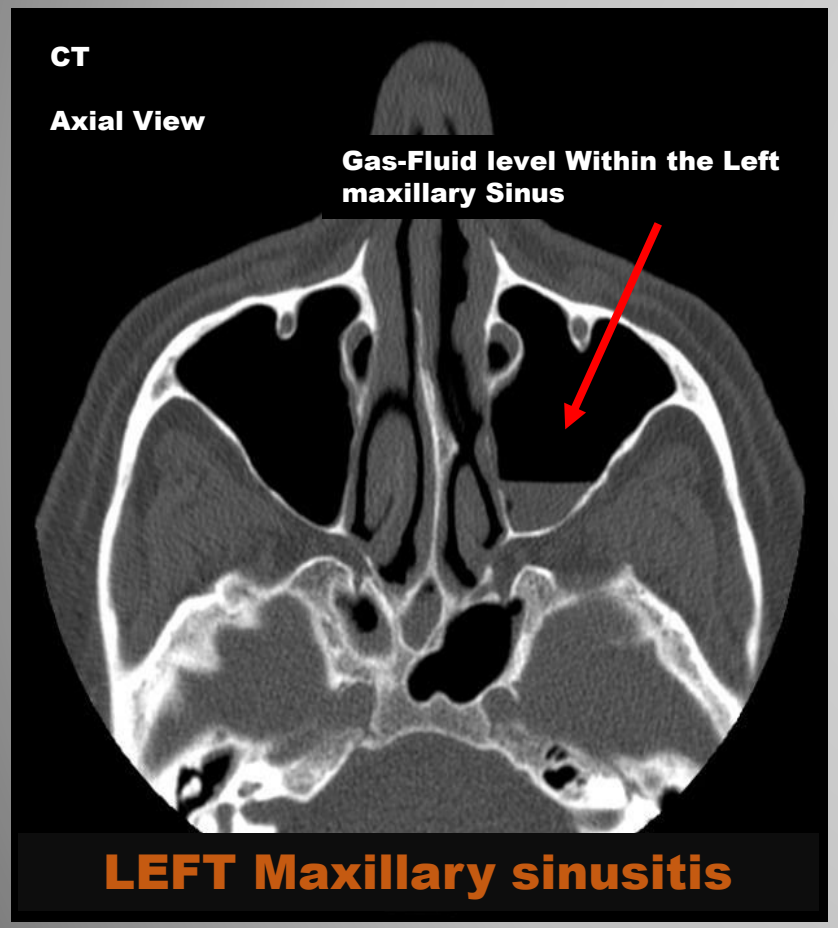
✓ **Peripheral mucosal thickening.**

✓ **Gas-fluid level.**

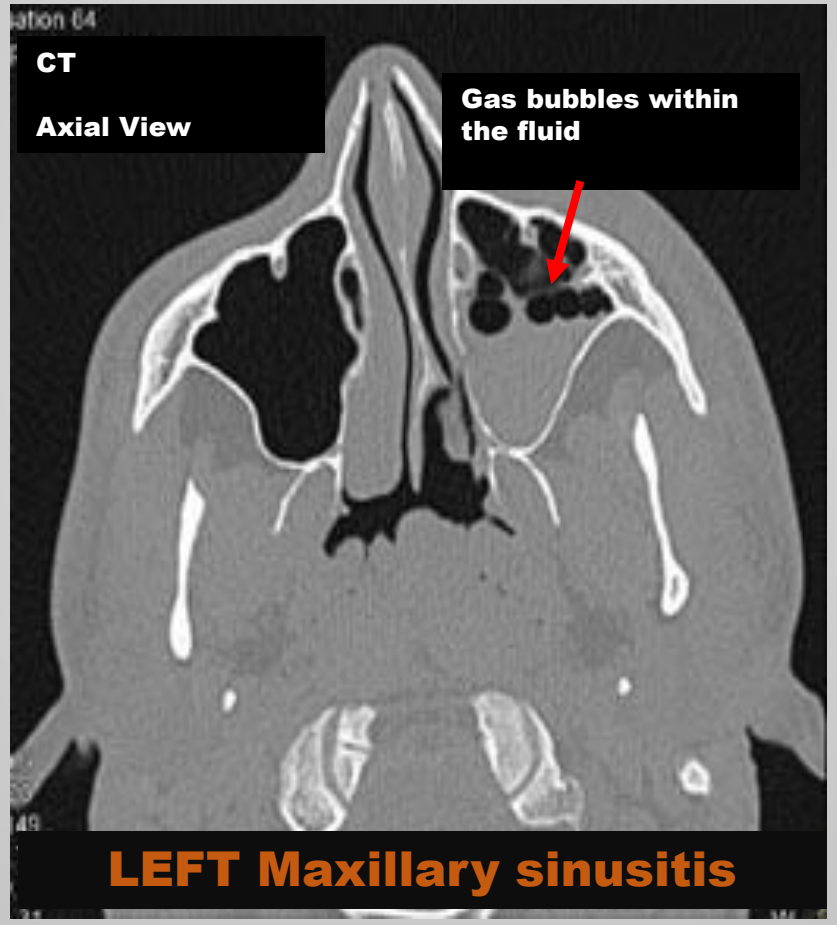
✓ **Gas bubbles within the fluid.**



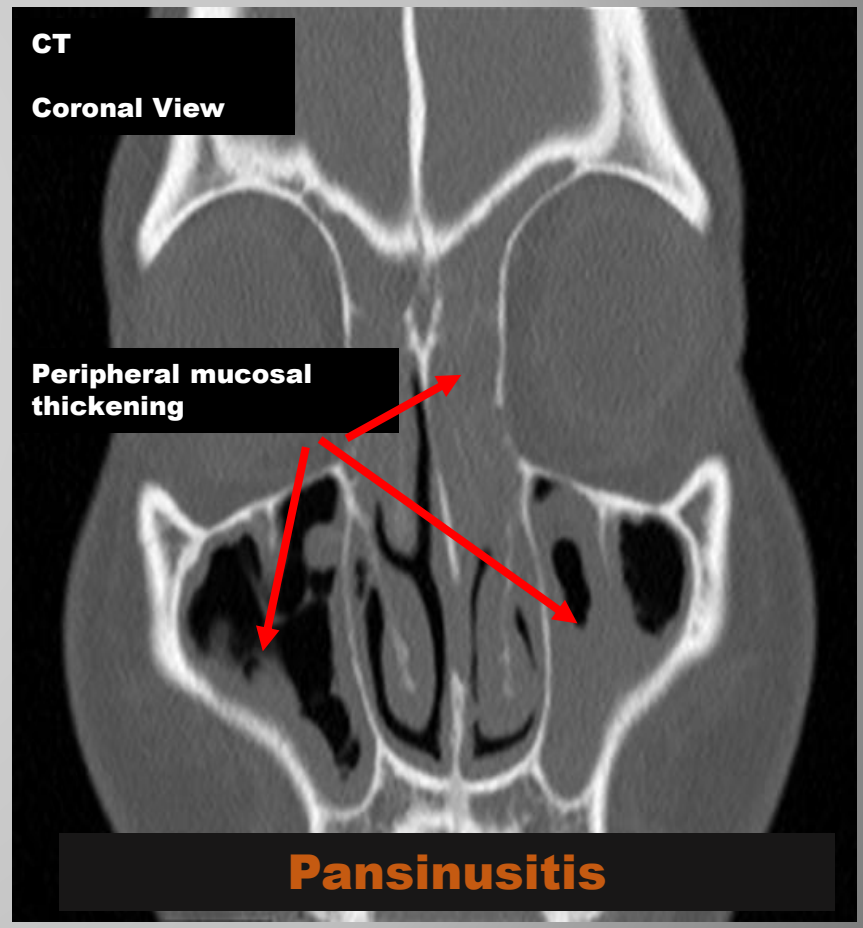
II. Acute Sinusitis



LEFT Maxillary sinusitis



LEFT Maxillary sinusitis



Pansinusitis

II. Acute Sinusitis

- **Complication:**

I. **Pott puffy Tumor.**

II. **Periorbital swelling.**

III. Dural venous sinus **thrombosis.**

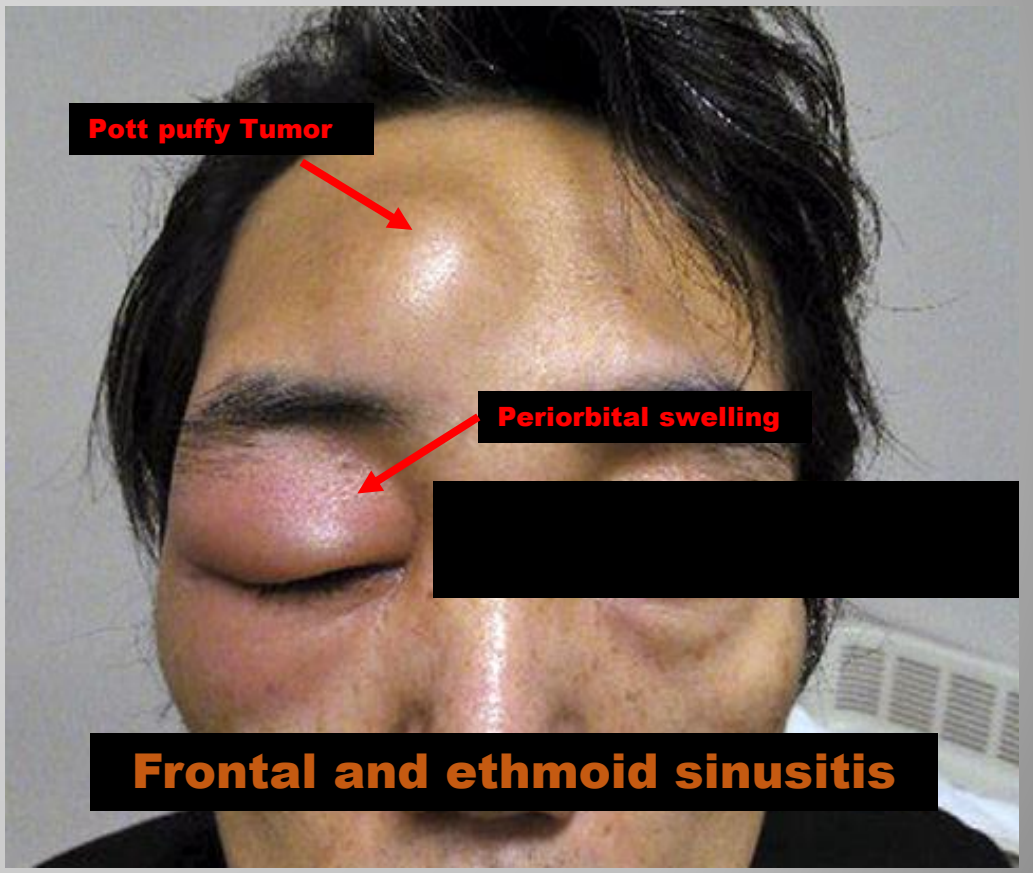
IV. Intracranial extension as **meningitis, subdural empyema and cerebral abscess.**

- **Treatment:**

A. Conservative medical treatment (including Antimicrobial) until the inflammation subside.

B. Treat the cause (Dental cares).

C. Surgery (complex cases)



III. Acute Epiglottitis

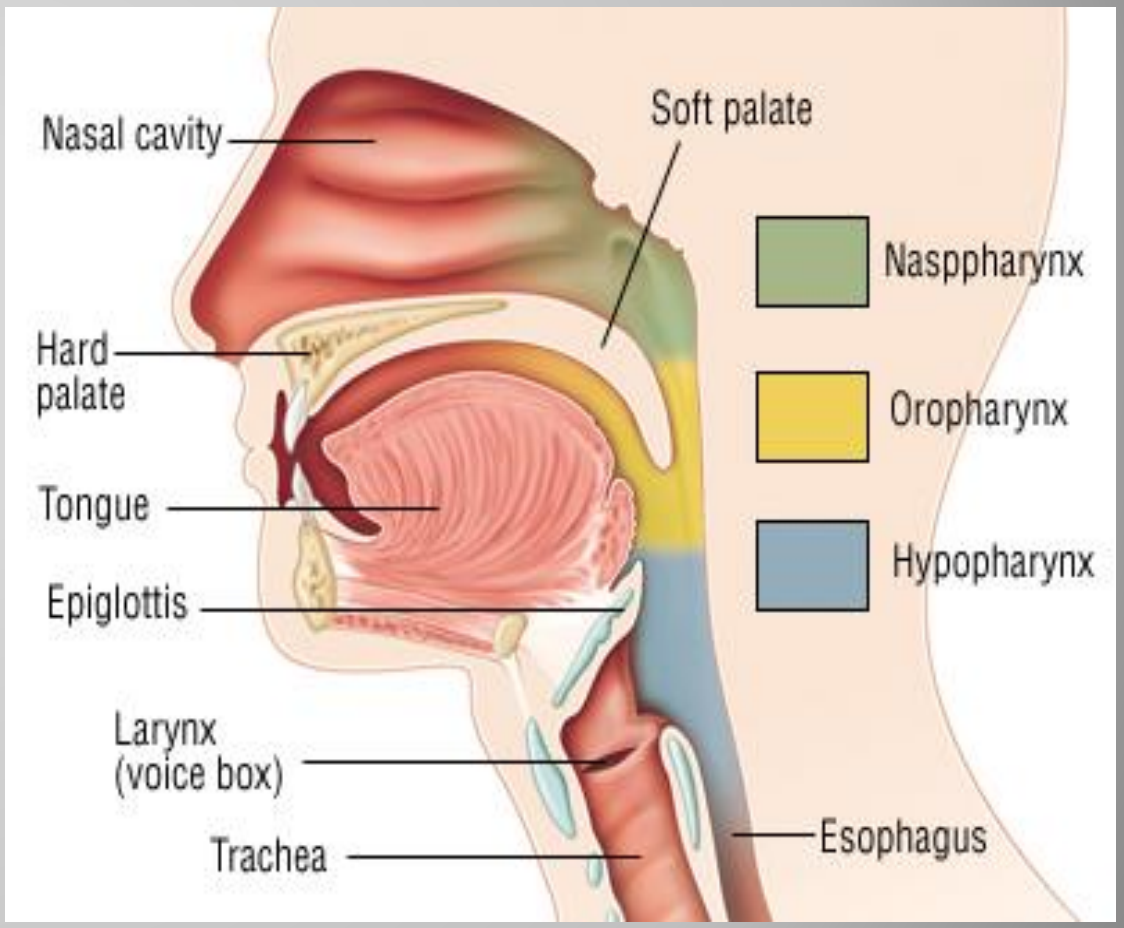
- **Definition:** **It is a life-threatening condition caused by inflammation of the epiglottis and aryepiglottic folds.**
- **It can lead to acute airway obstruction.**
- **Treatment is urgent.**

- **Age:** **3-6 years.**

- **Etiology:**

Haemophilus influenzae ***(most common)***

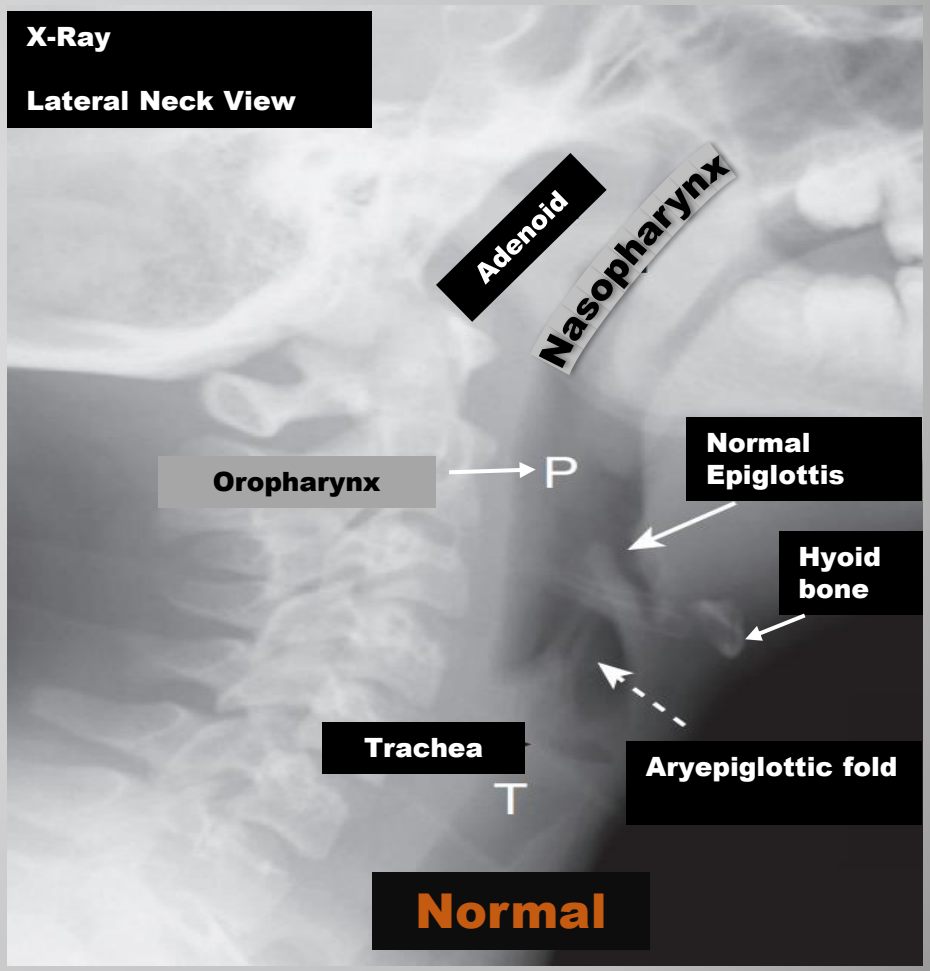
Group A beta-hemolytic streptococci



III. Acute Epiglottitis

- **Clinical presentation:**

1. Very rapid course beginning with a sore throat and leading to difficulty swallowing and drooling.
2. Fever
3. Difficulty speaking
4. Changes in the voice
5. Inspiratory stridor
6. Severe dysphagia



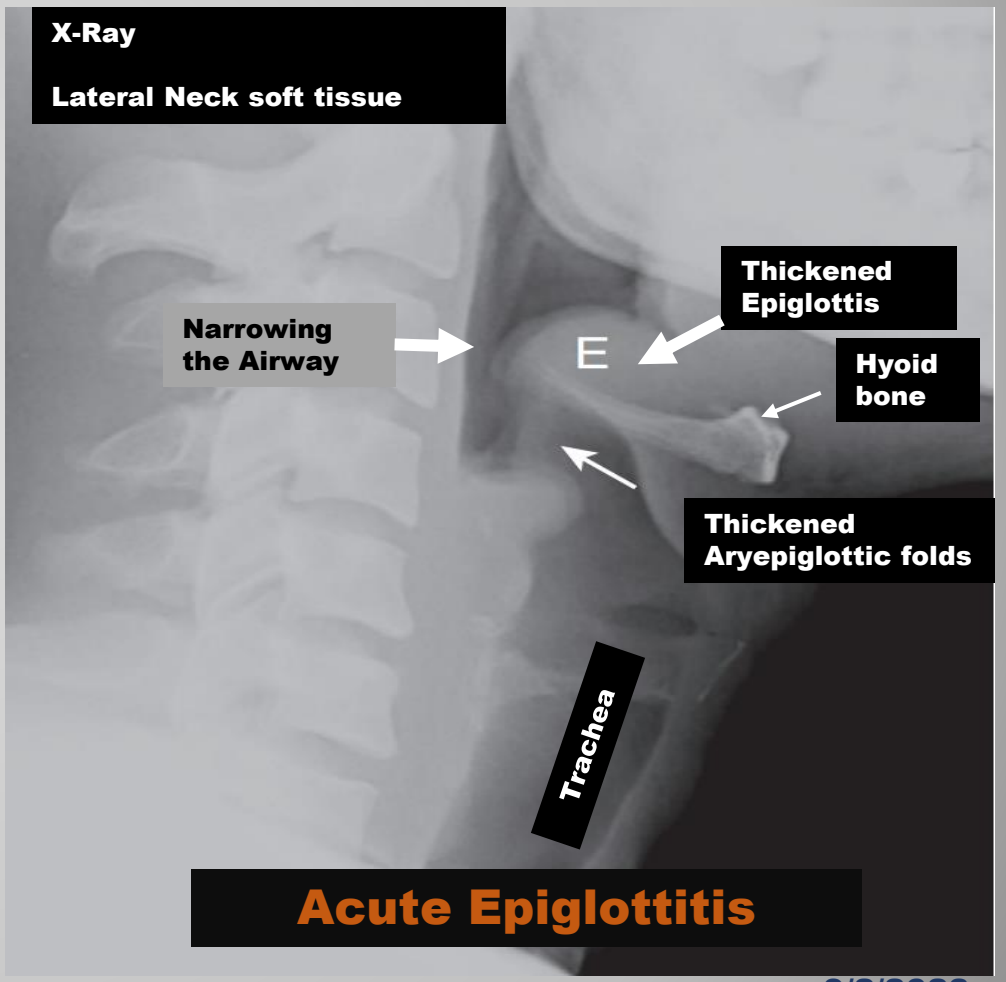
III. Acute Epiglottitis

- **Radiological manifestation:**

***X-ray (Lateral neck soft tissue Plain film):**
Thickening of the epiglottis and aryepiglottic folds referred to as the thumb sign.

Narrowing the airway.

Hypopharynx may be over-distended.



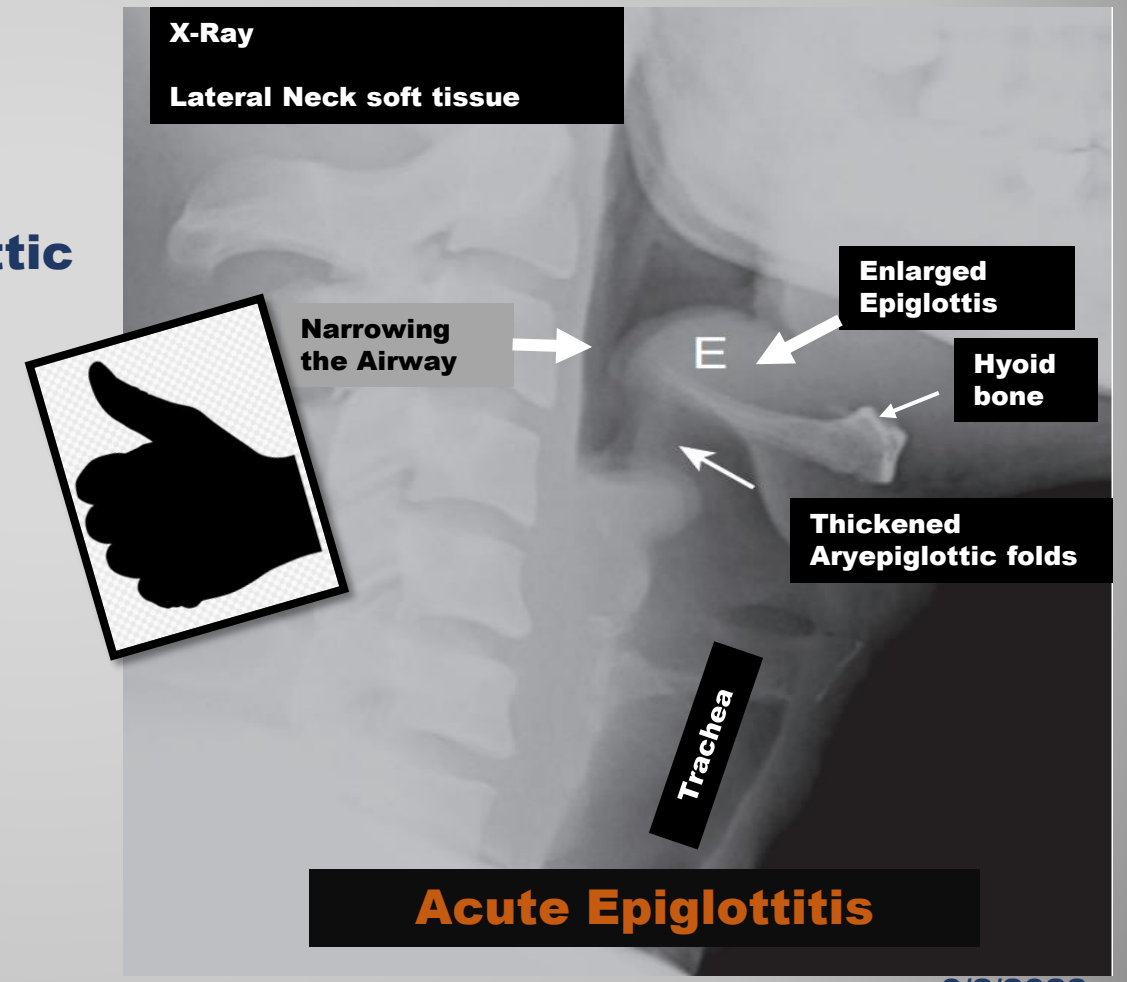
III. Acute Epiglottitis

- **Radiological manifestation:**

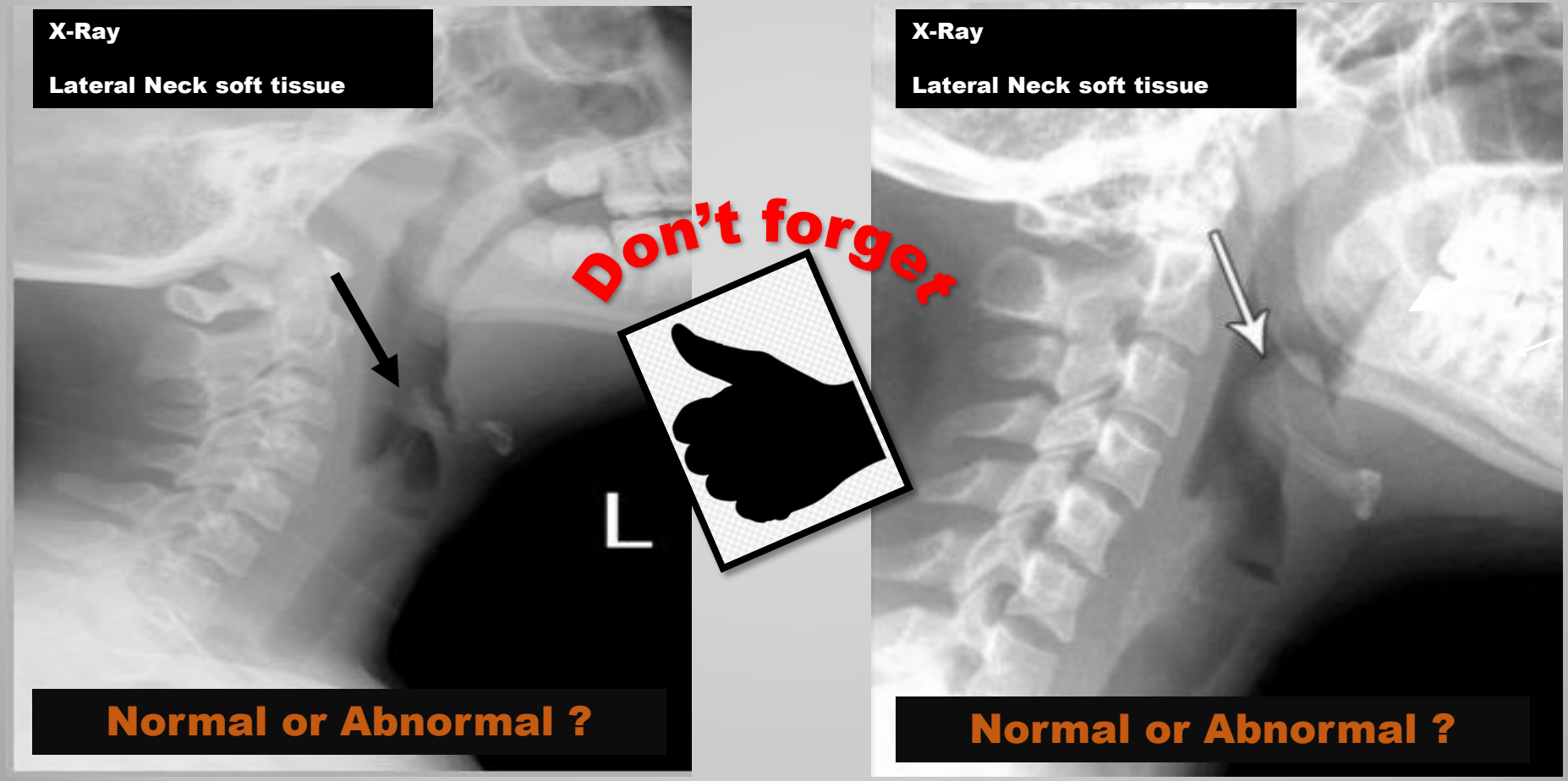
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III. Acute Epiglottitis



III. Acute Epiglottitis

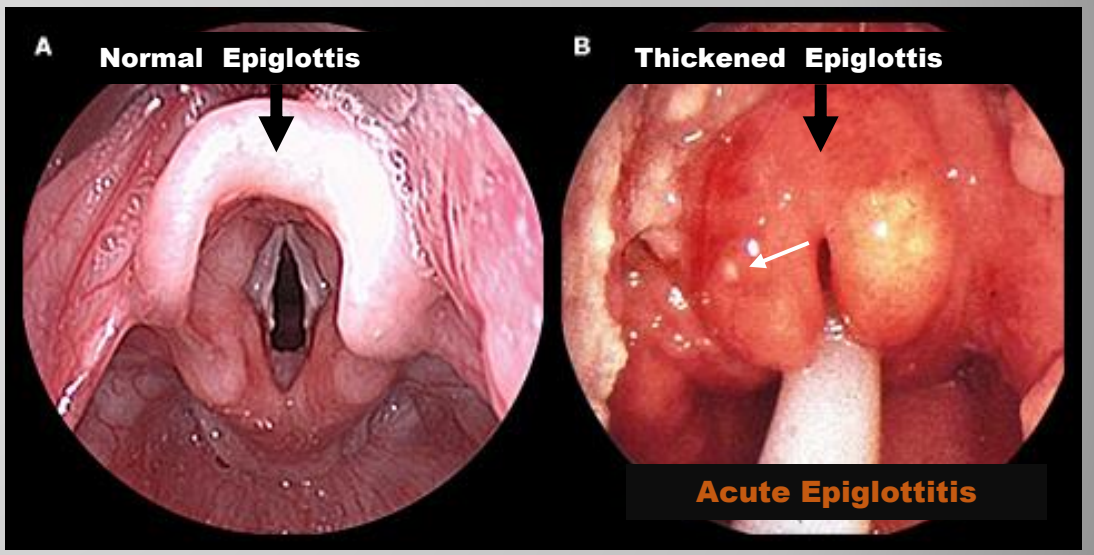
- **Treatment and prognosis:**

I. **Patients should be kept upright in a comfortable position.**

II. **Airway management with oxygen therapy.**

III. **Intravenous fluids.**

IV. **Intravenous steroids and antibiotics.**



Questions ?